Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.TECH. CE and IIIth
Name of the Paper: Fluid Mechanics
Paper Code: TCE-301
Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.
(iii) Please specify COs against each question.

Q1. (10 Marks)

a. Define the following terms:

- Newtonian Fluid
- Rheopetic Fluid
- Thixotropic Fluid
- Dilatant Fluid

(CO1)

OR

b. Classify the fluid.

du/dy	0	0.5	1.1	1.8
Shear Stress (τ)	0	2	4	6

(CO1)

Q2. (10 Marks)

a. Discuss the derivation of Bernoulli's equation from Euler's equation and explain its significance in fluid mechanics. (CO2)

OR

b. Explain the concept of buoyancy. Describe the factors that influence the stability of floating and submerged bodies with neat sketch. (CO2)

Q3. (10 Marks)

a. Discuss the Velocity potential function (ϕ) and Stream function (Ψ) in fluid kinematics. (CO2)

OR

b. Find the circulation around a closed curve defined by y=1, x=1, y=4 and x=4. When velocity held is given by U= 16y- 8x, and V= 8y-7x. (CO2)

Q4. (10 Marks)

c. Explain the term Viscosity. Discuss the variation of Viscosity with temperature and Pressure. (CO1)

Mid Term (Odd) Semester Examination October 2024

OR

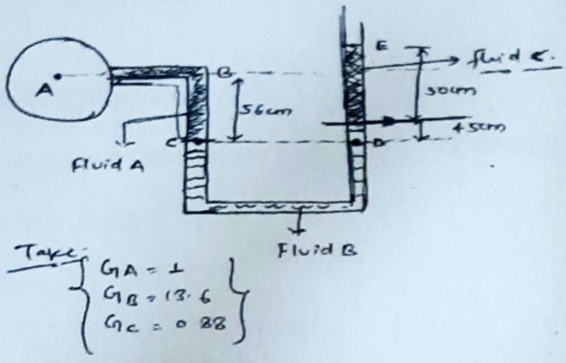
d. Explain the concept of hydrostatic paradox. Derive the equation of hydrostatic Law for plan surface. (CO1)

Q5. (10 Marks)

a. Define the following term:

- Gauge Pressure
- Absolute Pressure

Also Calculate the Pressure at A in term of head of Water as shown in figure. (CO1)



OR

b. Explain the term Path line. A fluid particles velocity vector is given by V= 2yi + 3xj, then find the acceleration of the particles at point (1,1) in X- direction. (CO2)